

CampusXchange: A Campus-Verified Marketplace for Selling, Lending, and Exchange

Project Proposal for UCS503P

Submitted to: Anushka Ma'am

Thapar Institute of Engineering and Technology

Garav Goyal, CSED

Parth Zutshi, CSED

Dhruv Rajput, CSED

Harjap Singh Benipal, CSED

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1 Higher-order goal

... secondary framing

This project aims to improve trust-based resource sharing within a campus community by reducing the friction, uncertainty, and safety gaps in how students currently buy, sell, lend, and exchange everyday items. While the underlying need is broader than any single campus, the immediate engineering objective is to translate *safe, verified peer-to-peer commerce* into a measurable, deployable software solution.

2 Time-to-value

... primary heuristic

- We will deliver meaningful increments quickly by building the Sell module first (the simplest complete transaction loop) and validating it with real users within a short iteration window.
- We will prioritize fast feedback over perfection by using weekly iterations (and targeting CI-backed automation early) to reduce release friction.
- Success is defined by what can be delivered and measured in the first iteration, then improved based on user feedback.

3 Problem Statement

Campus commerce today runs on ad hoc, trust-optional channels — hostel WhatsApp groups, batch chats, and a noticeboard here and there — that were never designed for hundreds of people who mostly don't know each other to buy, sell, lend, or swap items. The recurring pain points are:

- **No single place to look:** a listing might live in one hostel-wing group, one batch chat, or a physical board, so it rarely reaches anyone outside the poster's own circle — buyers and sellers who would gladly trade simply never find each other.
- **Strangers, no proof:** anyone can be added to these groups, including people with no real link to the college. There is no reliable way to confirm the other party is even a current student before money or an item changes hands.

- **Lending has no memory:** every exchange is treated as a one-off sale. Nothing tracks who borrowed what, when it's due back, or whether a deposit was returned — so lending barely happens at all, even though it's the natural fit for short-term-use items like calculators, lab coats, and textbooks.
- **No recourse when it goes wrong:** if an item arrives damaged, a seller vanishes after payment, or a borrowed item never comes back, there is no escrow, no verified handoff record, and no one to appeal to.

Why this matters now (contextual relevance): A campus effectively re-runs the same shopping list every semester — the same textbooks, calculators, and lab coats change hands within a fixed, repeatedly-renewing population. That repetition means even a modest fix to trust and discoverability compounds across thousands of small trades a year, in a way it wouldn't for a marketplace serving one-off strangers.

4 Proposed Solution

... Software Proposition

4.1 Overview

Build a campus-verified mobile marketplace app, “CampusXchange”, with the following components:

- **Verified access:** sign-up gated by a college email address (must end in `@thapar.edu`) plus each student's Thapar ID (roll number), entered by hand and checked for a valid format — no physical ID scanning or third-party document-verification vendor involved.
- **Sell, Lend/Borrow, and Exchange modules:** photo-first listings with price, condition, or rent terms; category-matched exchange requests.
- **In-app chat and negotiation:** real-time messaging and offer cards between buyer/borrower and seller/lender.
- **In-app payments:** a shared, escrow-style transaction core used by all three modules.
- **QR-verified handoff:** a dynamic, transaction-scoped QR code confirms the physical meetup before funds are released.
- **Trust and safety layer:** AI-assisted content moderation with mandatory human review, plus admin-adjudicated dispute handling.

4.2 Core workflow

1. Student creates a listing (sell, lend, or exchange) and receives a listing ID immediately.
2. System checks completeness and routes the listing to a moderation queue for human review before it goes live.
3. Interested students discover the listing, negotiate over in-app chat, and agree on terms (price, or rent duration and deposit for lend/borrow).
4. Both parties confirm the transaction; payment is held in escrow and a dynamic, time-bound QR code is issued.
5. At the in-person handoff, the QR is scanned and condition photos are captured; escrow is released, or the transaction is flagged for admin review if a discrepancy is reported.

4.3 Operational constraints for a single-campus pilot setting

- Scope the pilot to one college end-to-end: sign-up accepts only the @thapar.edu domain, meeting-location suggestions are pre-seeded with on-campus landmarks (hostels, academic blocks, main gate), and there is no cross-campus listing visibility at MVP.
- Design around semester-driven load rather than steady traffic: expect sharp spikes at the start of semester (Sell/Exchange for used textbooks and lab gear) and around move-out (graduating seniors clearing belongings), with quieter stretches mid-semester.
- Build for the phones and networks students actually have: a lightweight cross-platform app that stays usable on budget Android devices and patchy hostel/campus Wi-Fi, rather than assuming high-end hardware or constant connectivity.
- Keep every “smart” piece — moderation flagging, exchange matching — heuristic- and API-assisted rather than a custom-trained model, so effort goes into transaction correctness and workflow, not ML research.
- Meetups stay physical and on-campus by design, with no shipping or courier integration — which means the QR handoff and meeting-point selection are the entire physical-safety surface the project needs to get right.

5 Solution Approach

... *Engineering focus*

This is an engineering course project, so we focus on scalable marketplace workflow and trust-and-safety mechanisms rather than novel algorithm research.

5.1 Moderation and verification assistance

... *non-research*

Content moderation and identity checks will lean on simple rules and ready-made external services, not any custom-built AI model:

- Use an off-the-shelf vision/text moderation API to assign listings a confidence-scored flag, without claiming state-of-the-art detection.
- Verify identity using the student’s Thapar ID (roll number) entered at sign-up: check that it is a valid format, and, where a registrar-provided student list is available for the pilot, cross-check it against that list — no barcode scanning or OCR of a physical ID card is required.
- Show flagged listings and disputed transactions to a human moderator for decision; never fully auto-reject or auto-ban.

5.2 Mobile and backend architecture

... *mobile-based solution*

A typical mobile-plus-API architecture:

- **Frontend:** a cross-platform mobile app (listings feed, chat, transaction, and profile screens) with a separate web dashboard for admins.
- **Backend API:** authentication, listing management (create/edit/remove), matching/notifications, chat, and the shared transaction (escrow and QR) core.
- **Database:** store users, listings, chats, transactions, escrow holds, QR tokens, condition photos, and disputes.

5.3 CI/CD and fast delivery enabler

The project already commits, in Section 2, to CI-backed automation as part of its time-to-value heuristic — with a full-stack mobile client, backend API, and a payments/escrow path in scope, that automation earns its place rather than being boilerplate. CI/CD will be used to:

- Automatically build and run tests on every change.
- Produce repeatable deployment artifacts to staging.
- Enable safe and frequent releases of incremental features.

6 Evaluation Criterion

...measurable and attributable

We define success using measurable metrics tied to the core workflow, chosen so each one traces back to a pain point from Section 3.

6.1 Primary evaluation metric

Time-to-listing-clearance (TLC): time between a listing’s submission and its moderation decision (published or rejected) appearing in the system — a check that verified, moderated access (Section 3: “strangers, no proof”) doesn’t become a new bottleneck of its own.

- Target: median $TLC \leq 4$ hours during pilot window.
- Attribution: measured from database timestamps (submission vs. moderation decision by staff).

6.2 Secondary evaluation metrics

- **Transaction completion rate:** percentage of agreed transactions that reach a completed, QR-verified handoff — tests whether the escrow-and-QR safety net (“no recourse when it goes wrong”) actually closes the loop instead of stalling.
- **Lend/borrow share and on-time return rate:** share of transactions that are lend/borrow rather than sell, and the percentage of those returned by the agreed date without a dispute — tests whether native lending support (“lending has no memory”) gets used, not just built.
- **Dispute rate:** percentage of lend/borrow transactions flagged for admin review after condition-photo comparison.
- **Discovery over off-platform channels:** share of pilot users who report, via a one-question check-in, finding or listing an item in-app rather than falling back to WhatsApp or noticeboards — a proxy for whether fragmentation (“no single place to look”) is actually going down.
- **User clarity:** buyer/seller-reported clarity of transaction status using a simple 1–5 feedback question.
- **Reliability:** availability of staging/production for login, listing, and payment during peak hours (e.g., $\geq 99\%$ uptime in pilot).

6.3 Pilot validation plan

... *high-level*

- Recruit 10–20 pilot users across freshers, graduating seniors, and hostel residents, plus 2–3 volunteer moderators (or a simulated admin role if real staff are unavailable).
- Compare metrics before/after within the iteration window by logging baseline estimates via short interviews on current informal buying, selling, and lending habits.

7 Scalability

... *with foresight*

The proposition should scale in both theoretical and practical senses.

1. **Scalable (theoretically):** The system should support growth in number of listings and concurrent chats/transactions by:
 - decoupling components (mobile client/backend),
 - enabling pagination and indexing for listing and category search,
 - using stateless API design for the transaction core.
2. **Operationally deployable (practically):** The system should run on common cloud hosting:
 - managed database,
 - containerized or platform-hosted deployment,
 - CI/CD pipeline for staging/production.

8 Engine availability heuristic

A mature set of “engines” (tooling/services) is available to implement this as a mobile-based project:

- Cross-platform mobile framework, plus a REST/WebSocket API layer.
- Managed relational database, with an in-memory cache for sessions.
- Authentication approach (email OTP, token-based sessions) and simple roll-number/registrar-list cross-checking — no third-party KYC or document-verification vendor required for the pilot.
- Object storage for listing images and condition photos.
- Payment gateway with escrow/split-payment support.
- Test framework for unit/integration tests; CI runner and staging deployment target.

We will use these mature components to focus on marketplace workflow design, trust-and-safety correctness, and measurement rather than inventing new infrastructure.

9 Project scope and deliverables

... *iteration-friendly*

9.1 Initial deliverable

... *first iteration*

- Student auth (college email OTP + Thapar roll-number check) and basic Sell-listing management (create, edit, remove)
- Buyer chat/negotiation and seller-picks-buyer flow
- Basic in-app payment confirmation, with escrow logic stubbed initially
- Basic logging and timestamps for TLC measurement
- CI: automatic build and test runs on every code change
- CD to staging with a single command or pipeline job

9.2 Subsequent deliverables

- Lend/Borrow and Exchange modules
- QR-verified handoff and mandatory condition-photo capture
- AI-assisted moderation queue with human review
- Admin dashboard for metrics, moderation, and dispute resolution
- Improved search/filtering and indexed category queries for scale readiness

10 Risks and mitigations

- **Cold-start / liquidity:** seed the marketplace with orientation-week campaigns and incentives for early listers.
- **Off-platform leakage:** keep the in-app QR handoff simple and fast so there is little reason to move the transaction elsewhere.
- **Data privacy / consent:** minimize collected identity data — roll number only, no ID card image or scan; encrypt condition-photo storage; store only what is needed.
- **Dispute and damage adjudication ambiguity:** instrument mandatory before/after condition photos and define clear escalation and penalty tiers before development begins.
- **Operational issues in pilot:** use staging early; ensure CI/CD produces repeatable deployments.

11 Summary

This project proposes a deployable mobile marketplace to reduce fragmentation and safety gaps in campus buying, selling, and lending. It meets the course criteria by emphasizing time-to-value through rapid iterations, implementing CI/CD to support frequent safe releases, and using measurable evaluation criteria (especially time-to-listing-clearance). It remains focused on engineering workflow, trust-and-safety design, and scalability readiness rather than research-driven novelty.